

Extra — If Statements: Only Sometimes

Topic: If / Elif / Else · **Course:** Extra Worksheet · **Time:** about 30 minutes

Some code runs **every time**. Some code runs **only if** something is true. On paper you will read `if` code, predict it, fix it, and explain it in your own words.

Keep these words handy: **condition**, `==`, **else**, **elif**, **indent**.

1 · Predict 🧠

Read each snippet. Write what you think it will print.

```
score = 7
if score > 5:
    print("You win!")
```

Does "You win!" print? Why?


```
age = 10
if age == 12:
    print("You are 12.")
else:
    print("You are not 12.")
```

What prints?


```
weather = "rain"
if weather == "sun":
    print("Wear sunglasses")
elif weather == "rain":
    print("Bring umbrella")
else:
    print("Just go outside")
```

What prints? Why did it skip the first line?

2 · Spot the Bug 🐛

Each snippet is broken. Read what it should do, then rewrite it so it works. Explain why the original was wrong.

Bug A — This should check if `score` equals `10`. Instead it always prints "Perfect!", even when `score` is `3`.

```
score = 3
if score = 10:
    print("Perfect!")
```

Hint: checking equal uses two of something.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — This should print "Too cold" only when `temp` is below `0`. Right now it prints "Too cold" no matter what `temp` is.

```
temp = 15
if temp < 0:
    print("Too cold")
```

Hint: the line inside the `if` needs to be pushed in.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Explain the Code

Here is a complete program that checks a player's score and prints a message.

```
score = 8

if score >= 9:
    print("A+ player!")
elif score >= 5:
    print("Good job!")
else:
    print("Keep practicing!")
```

What prints when **score** is **8** ? Walk through why it picks that branch.

What would print if **score** was **9** ? What about **2** ?

Why does the program stop checking once it finds a true condition?

Submit 

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